Ofqual QAN	**600/0406/X / 600/0406/X**
Quartz ID	**1053 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing with Business Management**

Assessment	**Object-Oriented System Analysis and Design**
Assessment Type	**Global Examination**
Date	**Thursday 5th June 2025**
Time	**09.00am – 12.00pm**

Examination Paper

Answer ALL questions.

Clearly cross our surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

a) Define the following terms in the context of Object-Oriented Analysis and Design and provide an example:

i) Class attributes **2**

ii) Method **2**

iii) Abstract class **2**

b) Please select ONE (1) from the options provided for each statement below.

i) The bundling of data attributes and methods together within a single class to ensure the internal state of an object is kept private is called: **1**

A. Encapsulation
B. Refactoring
C. Abstraction
D. Inheritance

ii) In the SOLID design principles, the requirement for objects of a superclass to be replaceable by objects of its subclasses without affecting the correctness of a program is called: **1**

A. Single Responsibility Principle
B. Execution of Individual Methods Principle
C. Liskov Substitution Principle
D. Dependency Inversion Principle

iii) In basis path testing, a node that contains a condition is called: **1**

A. Edge
B. Predicate node
C. Region
D. Independent path

iv) In object-oriented design, when a component has a single, well-defined responsibility, this is referred to as: **1**

A. Cohesion
B. Coupling
C. Inheritance
D. Prioritisation

Total 10 marks

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Marks

Question 2

Scrum and Kanban are both Agile methodologies used for managing projects, but they differ in several key aspects. Describe FIVE (5) ways in which SCRUM differs from Kanban. **10**

Total 10 marks

Question 3

a) Draw and label a diagram of the Waterfall model, clearly illustrating the stages. **7**

b) Explain THREE (3) advantages of the Waterfall model. **3**

Total 10 marks

Question 4

Draw a Unified Modelling Language (UML) Use Case Diagram to illustrate the activities shown in the Use Case Matrix below. When doing this, demonstrate the key concepts of generalisation, <<include>>, <<extend>>. **10**

	Customer	**Orders clerk**	**Warehouse**
Place order • Enter customer details or call up registered customer details • Select items • Choose shipping preference • Submit order	X	X	
Change shipping details	X	X	X
Make Payment • By card • Cash payment • On account	X		

Total 10 marks

Question 5

a) Identify FIVE (5) metrics that can be used to assess system reliability. **5**

b) Identify FIVE (5) techniques that can be applied in pursuit of zero defects. **5**

Total 10 marks

Question 6

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Marks

Explain FIVE (5) advantages **and** FIVE (5) disadvantages of component re-use. **10**

Total 10 marks

Question 7

a) Define the term Testability in relation to software development. **2**

b) Explain FOUR (4) ways in which software testability can be assessed. **8**

Total 10 marks

Question 8

a) Compare **and** contrast Coupling and Cohesion providing ONE (1) example of each. **6**

b) Explain what is meant by information hiding. Provide an example to help explain your answer. **4**

Total 10 marks

Question 9

a) Identify THREE (3) advantages of prototyping. **3**

b) Describe the evolutionary prototyping method and explain THREE (3) advantages of applying evolutionary prototyping. **7**

In your answer you should clarify how evolutionary prototyping resembles or differs from throwaway prototyping or incremental prototyping.

Total 10 marks

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Marks

Question 10

Study the following description of part of a system:

The system records information about students, tutors and units that they study in a university course.

- Students study SIX (6) units and a unit can be studied by many students.
- A unit has a code, name and duration. A unit may have many presentations, defined by a start date and the tutor who teaches it.
- Students register for a particular unit presentation. A tutor may teach many unit presentations.
- The system should record the name and university id for each student and tutor and also the course code on which the student is registered.

a) Identify FOUR (4) candidate classes from the description. **4**

b) Draw a UML Class diagram showing how inheritance can be represented between classes in the system described above. You need not show attributes in your diagram. **3**

c) Draw a UML Class diagram showing an association between TWO (2) classes that you identified in Question 10a from the system described above. You should include the multiplicities in your diagram. You need not show attributes in your diagram. **3**

Total 10 marks

END OF PAPER

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Learning Outcomes Matrix

Task	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	2	Yes
2	1	Yes
3	1	Yes
4	2, 3	Yes
5	4	Yes
6	1, 4	Yes
7	1, 4	Yes
8	2	Yes
9	1	Yes
10	2, 3	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
1. Discuss various approaches to systems analysis and design and explain their strengths and weaknesses.	Unable to identify and assess approaches.	In an extremely limited way, can identify and assess methodologies and methods to reach limited solutions.	Can adequately identify and assess appropriate methodologies and methods to reach appropriate solutions.	Can sufficiently identify and assess appropriate methodologies and methods to reach good solutions.	Can coherently identify and assess coherent methodologies and methods to reach well-defined and structured solutions.
2. Apply object-oriented techniques for systems analysis and modelling of well-defined but complex problems.	Unable to explain key principles or discuss developments.	In an extremely limited way, can explain principles and discuss developments.	Can adequately explain principles and discuss developments.	Has very good grasp of explain principles and discuss developments.	Can coherently discuss developments and explain and adapt principles.
3. Explain and use UML to represent object-oriented models.	Little to no understanding of UML or object-oriented models.	Limited understanding of UML or object-oriented models.	Adequate ability to apply UML with sufficient representations.	Demonstrates very good ability to apply and adapt UML models with a substantial system understanding.	Outstanding ability to apply UML and adapt models in a manner that demonstrates understanding of effectiveness of methods, actions and results.
4. Understand the integration between the design models and implementation.	Lacks awareness of different perspectives or approaches within the area of study.	Has extremely limited awareness of different perspectives or approaches within the area of study.	Has adequate awareness of different perspectives or approaches within the area of study. Understands requirements traceability but with some gaps.	Has very good, informed awareness of different perspectives or approaches within the area of study. Understands requirements traceability.	Excellent awareness of different perspectives or approaches within the area of study.

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